

Complements of Operational Research

2.2 Project planning: PERT

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When durations are uncertain

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$$t_e = \frac{t_o + 4t_m + t_p}{6}, \quad \sigma^2 = \left(\frac{t_p - t_o}{6} \right)^2$$

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For B: $t_e = \frac{2 + 4 \times 3 + 10}{6} = 4$ weeks,

$$\sigma^2 = \left(\frac{10 - 2}{6} \right)^2 \approx 1.78.$$

The long tail (10) pulls the expected time above the most likely one (3).

The laboratory with three estimates

Act.	Pred.	t_o	t_m	t_p	t_e	σ^2
A	—	2	3	4	3	0.111
B	A	2	3	10	4	1.778
C	A	1	2	3	2	0.111
D	C	2	3	4	3	0.111
E	C	1	2	3	2	0.111
F	B, D, E	2	3	4	3	0.111
G	D	1	2	3	2	0.111
H	F, G	1	2	3	2	0.111

The expected times t_e are the durations of 2.1. Hence, with t_e :

- expected duration: 13 weeks;
- critical path: A, C, D, F, H.

What is new: each activity now has a **variance**. That of the equipment (B) is 16 times larger than the others.

PERT assumptions

1. The project duration is that of the critical path (computed with t_e).
2. The durations are independent and their sum is **approximated** by a normal (with few activities, it is only an approximation):

$$T \sim N\left(\mu = \sum_{\text{crit.}} t_e, \sigma^2 = \sum_{\text{crit.}} \sigma_i^2\right).$$

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In the laboratory: $\mu = 13$,
 $\sigma^2 = 5 \times 0.111 = 0.556$, $\sigma = 0.745$.

Question: probability of finishing in 14 weeks or less?

$$\begin{aligned} P(T \leq 14) &= P\left(Z \leq \frac{14 - 13}{0.745}\right) \\ &= P(Z \leq 1.34) \approx \mathbf{0.91}. \end{aligned}$$

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Question: and do we believe this 91%?

PERT only looks at one path

Path	μ	σ^2
A-C-D-F-H	13	0.556
A-B-F-H	12	2.111
A-C-D-G-H	12	0.556
A-C-E-F-H	12	0.556

A-B-F-H is shorter on average, but much more uncertain (because of B):

$$P(A-B-F-H \leq 14) = P\left(Z \leq \frac{14-12}{\sqrt{2.111}}\right) \approx 0.92.$$

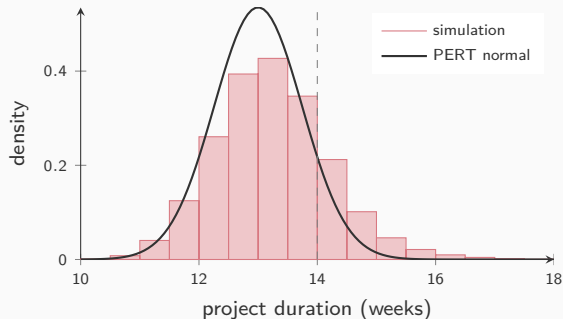
The project only finishes on time if **all** paths finish on time.

What PERT ignores

Near-critical paths with a large variance can delay the project, even though they are not the critical path computed with t_e . That is why PERT tends to be **optimistic**.

How can the effect be measured? **By simulation**: draw the durations of all activities many times and compute, in each draw, the project duration with the CPM of 2.1.

Simulation: 200 000 projects



Durations drawn from Beta-PERT distributions with the same t_o , t_m , t_p .

	PERT	simulation
mean duration	13.0	13.2
$P(T \leq 13)$	50%	41%
$P(T \leq 14)$	91%	80%
$P(T \leq 15)$	99.6%	96%

In 23% of the simulations the longest path was A–B–F–H, and not the “critical” one.

With 14 weeks, the risk of delay is twice what PERT indicates.

On the computer: the simulation in Python

```
import numpy as np
rng = np.random.default_rng(2026); N = 200_000
# (predecessors, to, tm, tp)
P = {'A':([],2,3,4), 'B':(['A'],2,3,10), 'C':(['A'],1,2,3),
     'D':(['C'],2,3,4), 'E':(['C'],1,2,3),
     'F':(['B','D','E'],2,3,4), 'G':(['D'],1,2,3),
     'H':(['F','G'],1,2,3)}
def beta_pert(a, m, b):          # Beta with mean (a+4m+b)/6
    al = 1 + 4*(m-a)/(b-a); be = 1 + 4*(b-m)/(b-a)
    return a + (b-a)*rng.beta(al, be, N)
EF = {}
for k, (pre, a, m, b) in P.items(): # topological order
    ES = np.max([EF[p] for p in pre], axis=0) if pre else 0
    EF[k] = ES + beta_pert(a, m, b)
T = EF['H']
print(T.mean(), (T <= 14).mean())    # ~13.2 ~0.80
```

It is the CPM of 2.1 (forward pass), applied to 200 000 projects at once.

The same can be done in Excel (one row per simulation) or in MATLAB (betarnd).

Simulation does not require normality or independence between paths; it can also incorporate dependencies between the activity durations, if these are modelled (this code draws them independently).

PERT example (durations in weeks)

Activity	Predecessors	t_o	t_m	t_p
A	—	5	6	7
B	—	1	3	5
C	—	1	4	7
D	A	1	2	3
E	B	1	2	9
F	C	1	5	9
G	C	2	2	8
H	E, F	4	4	10
I	D	2	5	8
J	H, G	2	2	8

- Build the AON network.
- Build the AOA network.
- Determine the critical path and the time needed to complete the project.
- What is the probability of completing the project in 22 weeks?

Food for thought: is there any near-critical path with a large variance?

KEY TAKEAWAYS — 2.2

1. Three estimates: $t_e = (t_o + 4t_m + t_p)/6$, $\sigma^2 = ((t_p - t_o)/6)^2$.
2. With the t_e the CPM is done: expected duration and critical path.
3. $T \approx N(\sum t_e, \sum \sigma^2)$ **on the critical path**; $P(T \leq t) = P(Z \leq \frac{t-\mu}{\sigma})$.
4. PERT ignores near-critical paths with a large variance: it tends to be **optimistic**.
5. **Simulation** (CPM repeated with drawn durations) gives the probability without those assumptions.

How to show the plan on a calendar, and what does speeding it up cost? → 2.3 Gantt and *crashing*.

Exercise sheet 2 — Exercise 2 (homework)

Launching a mobile app (durations in weeks)

	Activity	Pred.	t_o	t_m	t_p
A	Specification	—	2	3	4
B	Interface design	A	2	4	6
C	Server (<i>backend</i>)	A	4	6	14
D	App (<i>frontend</i>)	B	3	5	7
E	Integration	C, D	1	2	3
F	Testing	E	1	2	9
G	Marketing campaign	A	2	3	4
H	Launch	F, G	1	1	1

- a) Build the AON network.
- b) Compute t_e and σ^2 for each activity.
- c) Determine the critical path and the expected project duration.
- d) What is the probability of launching the app in 20 weeks or less?
- e) Is there any path that, while not critical, deserves attention? Justify.
- f) (Optional) Estimate the probability in d) by simulation.

Submission: by email to the lecturer, by the deadline given in class. All exercises: **Exercise sheet 2**.

References and resources

References Full list: *Course bibliography*.

- Hillier, F. S., Lieberman, G. J., *Introduction to Operations Research*, 11th ed., McGraw-Hill, 2021 — Project Management with PERT/CPM (including the simulation).
- Heizer, J., Render, B., *Operations Management*, Pearson — ch. 3, Project Management.

Resources

- Supplementary code: PERT simulation in Python (this deck); PERT exercise solved in MATLAB and Python (Moodle).
- Exercise sheet 2 — Exercises 2 and 6.